

$$3 + 3.5 + 3 = 9.5$$

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Thermodynamics, Final Exam, 2023-24

Write your results in the boxes, and justify your answer in the space provided by using brief sentences and summarized calculations. No results without justification will be accepted.

Problem 1 (3 points)

A cylinder made of adiabatic walls has an adiabatic piston. The piston is frictionless and has negligible mass. The cylinder and piston contain $N = 2$ mols of a monoatomic ideal gas. The surroundings are at constant pressure P_{ext} of 1 atm and temperature T_{ext} of 25°C. The piston sits vertically and a certain mass of a liquid has been placed over the piston, so the gas is initially at a pressure P_i of 1.2 atm. The gas initial temperature T_i is equal to T_{ext} . In a first stage (a), the liquid mass is removed slowly (reversibly) until it is all gone. In a second stage (b), the system is left alone, and eventually, the adiabatic walls lose their thermal-insulation effectiveness and the gas slowly (quasistatically) reaches thermal equilibrium with the surroundings.

1. Find the temperatures and pressures at the end of each stage: T_{fk} , P_{fk} ($k = a, b$). [1 point]

To do so, consider that a) is a reversible adiabatic expansion, and b) a heat exchange at constant pressure. For a), note that we know $\frac{V_i}{V_f} = \left(\frac{T_f}{T_i}\right)^{\frac{\gamma}{\gamma-1}}$, which, since we only know initial pressure is $P_i = 1.2$ atm and final is $P_f = 1$ atm, we need it as: $\frac{P_f}{P_i} = \left(\frac{T_f}{T_i}\right)^{\frac{\gamma}{\gamma-1}}$. Finally, to get the final temperature, we isolate T_f and get:

$$T_f = T_i \cdot \left(\frac{P_f}{P_i}\right)^{\frac{\gamma-1}{\gamma}} = 298.15 - \left(\frac{1}{1.2}\right)^{\frac{2}{5}} = 277.18 \text{ (Note that, since it is a monoatomic ideal gas } C_v = \frac{3}{2}R \text{ and } C_p = C_v + R = \frac{5}{2}R)$$

T_{fa} (K) =	277,18
P_{fa} (Pa) =	101325
T_{fb} (K) =	298,15
P_{fb} (Pa) =	101325

For b), when walls become diathermal, since the P_{ext} is const., this will be an isobaric heat exchange until reaching thermal equilibrium, that means no other that $T_{fb} = T_{ext} = 298.15$. And, since $P_{initial} = 1 \text{ atm} = P_{ext}$, it will yield $P_{fb} = 1 \text{ atm}$.

2. Compute the work W_k and the heat Q_k on the gas in each stage, as well as the enthalpy changes ΔH_k ($k = a, b$) [2 points]

For a), since it is an adiabatic process, note $\Delta U = Q + W$ where $Q = 0$ J. Then, note that since it is an ideal monoatomic gas: $\Delta U = \frac{3}{2}NR\Delta T = \frac{3}{2} \cdot 2 \cdot 8.314 \cdot (277.18 - 298.15)$

$$= -523.027 \text{ J} = W_a$$

For b), consider that it is heat exchange at $P = \text{cte}$.

Then note that $dQ|_{P=\text{cte}} = dH|_{P=\text{cte}}$. Therefore $dQ|_P = dH|_P \cdot dT = Nc_p dT$

$$Q_b = Nc_p \int_{T_1}^{T_2} dT = Nc_p (T_2 - T_1)$$

W_a (J) =	-523,027
Q_a (J) =	0
W_b (J) =	-348,68
Q_b (J) =	871,712
ΔH_a (J) =	-871,712
ΔH_b (J) =	871,712

$$\text{As for the } W_b, \quad T_2 = 298.15, \quad T_1 = 277.18$$

$$= \frac{5}{2} \cdot 2 \cdot 8.314 \cdot (298.15 - 277.18) = 871.712 \text{ J} = \Delta H_b$$

note $\Delta U = Q + W$

$$W = \Delta U - Q = \frac{3}{2}NR\Delta T - Q = -348.68 \text{ J} \quad (\Delta T = (298.15 - 277.18))$$

$$\text{For } \Delta H_a, \text{ note } \Delta H = \Delta U + \Delta(PV) = \Delta U + \Delta(NRT) = \frac{3}{2}NR\Delta T + NR\Delta T = \frac{5}{2}NR\Delta T = -871.71 \text{ J}$$

Problem 2 (4 points)

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Assume
monatomic
for the gas

A small quantity (3 mg) of a liquid substance of molar mass 32 g/mol, initially at room conditions of 25°C and 1 atm, is rapidly injected into a heated chamber of 100 mL. The chamber is initially empty and it is sealed after the liquid injection. All the time, it is in thermal contact with a constant temperature source, at 352 K. The vapor pressure P_v (mmHg) of the liquid satisfies: $\text{Log}(P_v) = A - B/(T + C)$, where Log is the base-10 logarithm, T is the absolute temperature, and $A=8.07$, $B=1575$, $C=-34$. (These are known as the Antoine equation and Antoine equation parameters, respectively). We assume that the vapor phase behaves like an ideal gas. The heat capacity of the liquid is assumed to be constant, at 84 J/(mol·K). C_p !

1. Provide the final equilibrium state of the substance in the chamber, by reporting its volume V_f , pressure P_f , and temperature T_f . [1 point]

To do this, note that, since the liquid injected is a very small amount, we will assume that it undergoes heating at constant pressure, considering it will expand but never reach the full capacity of the chamber (since it is a very small amount).
Then, to see if it reaches the coexistence curve, calculate $T = \frac{B}{A - \log(760)} - C = 337.52 \text{ K}$

Moles were
plotted through
the molar and
molar mass
 $N = 9.375 \cdot 10^{-5} \text{ mol}$

$$\begin{aligned} V_f (\text{m}^3) &= 0.0001 \\ T_f (\text{K}) &= 352 \\ P_f (\text{Pa}) &= 2743.62 \end{aligned}$$

Since it does, it will then make a transition into gas, and once it is all gas, it will occupy all the chamber, and due to it being ideal, final pressure will be $P = \frac{NRT}{V} = 2743.62 \text{ Pa}$ (Note final temperature will be $T = 352$, since it will be at thermal equilibrium only then)

2. What is the heat of vaporization of the liquid at the source temperature? [1 point]

For this, consider (Clausius-Clapeyron) which yields the form $\ln(P) = -\frac{\Delta H_v}{R} \cdot \frac{1}{T} + C$. Note that, with $\log_{10}(x)$ we need to apply:

$$\log_{10}(x) = \ln(x) - \log_{10}(e) \Rightarrow \text{then: } \log_{10}(P) = -\frac{\Delta H_v \log_{10}(e)}{R} \cdot \frac{1}{T} + C'$$

However, note that in the Antoine equation, we have

$$\Delta H_v (\text{kJ/mol}) = 30.151$$

$$\text{Then: } \Delta H_v = \frac{B \cdot R}{\log_{10}(e)} = 30.151 \text{ kJ/mol}$$

$$\Delta H_v = \frac{RT^2 B \ln 10}{(T+C)^2} = 36.9 \frac{\text{kJ}}{\text{mol}}$$

3. Ignoring the result from the previous question, we now assume that the heat of vaporization of the liquid is 50 kJ/mol. Compute the changes in entropy and enthalpy experienced by the substance through the whole process (between just before being injected, and the final equilibrium) [2 points]

To do so, consider it is a three-stage process: 1) isobaric heating until $T = 337.52$; Transition 2) From liquid to gas; 3) Reaching the final conditions so that $T_f = 352 \text{ K}$.

• Enthalpy:

$$\rightarrow \Delta H_1: \text{recall } N = 9.375 \cdot 10^{-5} \text{ mol. Then, since isobar } \Rightarrow \Delta H_1 = \Delta Q_1 = \int_{T_1}^{T_2} N c_p dT = N c_p (337.52 - T_i) = 0.31 \text{ J}$$

For the transition, we use:

$$\rightarrow \Delta H_2 = N \cdot \Delta H_{v, \text{molar}} = N \cdot 50000 = 4.6875 \text{ J}$$

$$\Delta H_3 = 0.028 \text{ J}$$

$$\text{Then: } \Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$$

$$\rightarrow \Delta H_3 \Rightarrow \text{since } \Delta H = \Delta U + d(PV) = \Delta U + \Delta(NRT)$$

$$\text{given at is a monatomic ideal gas } \Rightarrow \Delta H = \frac{5}{2} N R \Delta T$$

$$\Delta S (\text{J/K}) = 0.0149$$

$$\Delta H (\text{J}) = 5.026$$

• Entropy:

$$\Delta S_1 = \int \frac{dQ_{\text{rev}}}{T} = N c_p \int \frac{dT}{T} = N c_p \ln \left(\frac{T_2}{T_i} \right) = 9.766 \cdot 10^{-4} \text{ J/K}$$

→ For ΔS_2 , note that, since it is an isobaric and isothermal process

$$dS = \frac{dQ_{\text{rev}}}{T} = \frac{dH}{T} \Rightarrow \Delta S_2 = \frac{\Delta H_2}{T} = 0.0139 \frac{\text{J}}{\text{K}} \quad (337.52 \text{ K})$$

$$\rightarrow \text{Finally, } \Delta S_3 \Rightarrow \text{through } \Delta S = \int_{T_2}^{T_3} \frac{c_v dT}{T} + N R \ln \left(\frac{T_3}{T_2} \right)$$

• Then:

$$\Delta S = \Delta S_1 + \Delta S_2 + \Delta S_3 = 0.0149 \text{ J/K}$$

$$\Rightarrow \Delta S_3 = 4.91 \cdot 10^{-5} \frac{\text{J}}{\text{K}} \quad \frac{3}{2} N R \Delta T$$

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Problem 3 (3 points)

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We consider the process of sublimation of 50 g of solid CO_2 ('dry ice'; molar mass 44 g/mol) at the conditions of -78.5°C and 1 atm, taking into account that such a state lies on the solid-vapour coexistence curve, approximately given by the equation:

$$\ln P = 27.0978 - 3031 \frac{1}{T}; \quad \text{where } P \text{ is given in Pa and } \ln \text{ is the natural logarithm.}$$

1. Determine ΔG for this process [1 point].

Given that our conditions are such that we are at the coexistence line, consider that the process of sublimation is isothermal and isobaric. The variations for any component for its Gibbs Free energy are:

$$dG^{(i)} = -S_i dT^{(i)} + V_i dP^{(i)} + \mu_i dN^{(i)} \Rightarrow \mu_i dN^{(i)}$$

However, we studied that, at the coexistence curve, $\mu^s(T, P) = \mu^g(T, P)$, then:

$$dG = dG_1 + dG_2 = \mu_1 dN^{(1)} + \mu_2 dN^{(2)} \quad \text{where } dN = dN^{(1)} = -dN^{(2)}$$

$$\Delta G \text{ (J)} = 0$$

$$\text{and so } dG = (\mu_1 - \mu_2) dN = 0 \Rightarrow \Delta G = 0 \neq$$

2. Determine ΔF for this process, considering that the density of the solid CO_2 is 1.56 g/mL [1 point].

For ΔF , see that the Gibbs Free energy is a Legendre transformation that can be written as $G = F + PV$. Then, given isobaric: $\Delta G = \Delta F + P\Delta V \Rightarrow \Delta F = \Delta G - P\Delta V$.

To obtain ΔV , consider the change in volume from solid to gas.

$$\rightarrow \text{Solid: } 50 \text{ g } \text{CO}_2 \cdot \frac{1 \text{ mL}}{1.56 \text{ g}} \cdot \frac{1 \text{ m}^3}{10^6 \text{ mL}} = 3.205 \cdot 10^{-5} \text{ m}^3 \text{ (solid)}$$

$$\rightarrow \text{Gas: we assume ideal gas behaviour} \rightarrow V = \frac{NRT}{P} \rightarrow \text{moles: } 50 \text{ g } \text{CO}_2 \cdot \frac{1 \text{ mol}}{44 \text{ g}} = 1.136 \text{ mol}$$

$$V = \frac{1.136 \cdot 8.314 \cdot 194.65}{1.01325 \cdot 10^5} = 0.0181495 \text{ m}^3 \text{ (gas)}$$

$$\Delta F \text{ (J)} = -1835.753$$

$$\text{Then: } -P \cdot \Delta V = -101325 \cdot (V_{\text{gas}} - V_{\text{solid}}) = -1835.753 \text{ J}$$

3. Determine ΔH for this process [1 point].

Given the equation for the coexistence line and that the process is a solid-vapor transition, consider Clausius-Clapeyron:

$$\ln\left(\frac{P}{P_0}\right) = -\frac{\Delta H_v}{R} \left(\frac{1}{T} - \frac{1}{T_0}\right) \Rightarrow \ln(P) = -\frac{\Delta H_v}{R} \cdot \frac{1}{T} + \left(\ln(P_0) + \frac{\Delta H_v}{RT_0}\right)$$

Therefore, we have:

$$-\frac{\Delta H_v}{R} = -3031 \Rightarrow \frac{\Delta H_v}{R} = 3031 \cdot 8.314 = 25399.734 \text{ J/mol} \rightarrow \text{si multiplication pels } 1.136 \text{ mols}$$

$$\Delta H \text{ (J)} = 28636.06$$

$$\begin{aligned} &\Rightarrow \Delta H_v \cdot 1.136 \\ &= 28636.06 \text{ J} \\ &= \Delta H_{\text{total}} \end{aligned}$$